scientific reports

OPEN

Automated violence monitoring system for real-time fistfight detection using deep learning-based temporal action localization

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Fistfight detection in video data is a critical task in video surveillance systems, where identifying physical altercations in real-time can enhance safety and security in public spaces. Earlier techniques primarily emphasized capturing inter-person interactions and combining individual characteristics into group-based representations, often overlooking the critical intra-person dynamics within the human bodypose point framework. However, essential individual features can be extracted by examining human skeletal movements' progression and temporal patterns. This paper presents a novel multimodal spatio-temporal fistfight detection model (MSTFDet) that integrates RGB images and human skeletal data to identify violent behaviors accurately. The proposed framework leverages both Context-Aware Encoded Transformer (CAET) for modeling interactions between individuals and their environment and Spatial–Temporal Graph Convolutional Networks (ST-GCN) for capturing intra-person and inter-person dynamics from skeletal data. The RGB module uses a combination of spatial and temporal transformers to model contextual relationships and individual actions, while the bodypose-point module processes skeletal data to capture the fine-grained motion of individuals. We conduct evaluations on two public datasets: the Surveillance Camera Fight Dataset (SCFD) and the RWF-2000 dataset, which feature complex real-world scenarios. On the SCFD and RWF-2000 datasets, MSTFDet achieved a Multi-class classification accuracy (MCA) of 92.3% and 95.2% MCA, respectively. These results highlight the effectiveness of the proposed approach in capturing both spatial and temporal features, providing a robust solution for real-time fistfight detection in diverse and challenging environments.

Keywords Violence detection, Vision transformer, Deep learning, Action recognition, Multimodal

Fistfight detection within video data, as a specific form of violence recognition, involves identifying violent behaviors such as physical altercations in videos. This task is particularly important in various video surveillance applications, including environments like public transportation hubs, correctional facilities, and psychiatric centers. Unlike conventional video action recognition tasks, fistfight detection presents unique challenges due to the presence of multiple individuals and the complex, dynamic interactions between them[1]. Fistfight detection in video, as a specialized sub-task of video violence recognition, requires precise and thorough analysis of human motion. Accurately detecting violent fistfights necessitates understanding not only the physical properties of motion, such as direction and velocity, but also the contextual semantics of the movements[2]. Consequently, it becomes essential to combine contextual information from both the current environment and prior frames to fully interpret the ongoing motion[3]. Models capable of capturing spatial–temporal relationships across different regions of a video are employed to extract this type of information from 3D video data[4]. However, this task poses several challenges, including the complexity of spatial–temporal patterns, the duration of video sequences, the difficulty of assembling large-scale, comprehensive datasets focused on fistfight violence, and the redundancy of information often present in video datasets[5].

Convolutional models, which have traditionally dominated computer vision tasks, have been extensively utilized for violence recognition in videos, including fistfights[5,6]. Early approaches combined 2D convolutional

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backbones with recurrent feature aggregators to capture motion dynamics[7]. A significant advancement in video classification emerged with the introduction of 3D convolutional layers, which enabled the direct extraction of spatiotemporal features from video sequences[8]. By eliminating the need for recurrent aggregation, 3D convolutional backbones enhanced the stability of video model training[9]. Moreover, auxiliary features, such as optical flow[10], skeleton estimations[11], and deep features[12], were introduced to extract more valuable insights from video clips. Although CNN-based models for violence recognition have seen improvements in accuracy, challenges persist regarding their computational complexity and learning capacity, which limit their scalability and efficiency in real-time fistfight detection applications[2].

Vision Transformers represent a novel approach to capturing spatial-temporal information in 3D video data. Unlike CNNs, which rely on convolutional layers, Vision Transformers employ self-attention mechanisms to recognize patterns in video data[13]. The self-attention mechanism has the advantage of identifying long-range dependencies between image patches without heavily depending on multilayer architectures, as convolutional filters do. As a result, Vision Transformers are more efficient at capturing global features across video frames[14]. Additionally, the self-attention mechanism is more hardware-friendly, allowing modern deep-learning accelerators to handle computations more effectively, which is advantageous in the context of fistfight detection[2,15].

Accurately modeling the spatial-temporal relationships and dynamics of joints is essential for effective action recognition from bodypose sequences. In recent years, Graph Convolutional Networks (GCNs) have emerged as the dominant solution for this task, as they are well-suited for handling the irregular structure of human skeletal data[16]. By developing sophisticated graph topologies or traversal strategies, GCN-based methods have demonstrated significant improvements in recognition performance[17,18]. Concurrently, the rise of the Transformer architecture[19] has garnered substantial interest, leading to notable advances in various computer vision tasks[20–22]. For skeleton-based action recognition, the self-attention mechanism inherent in the Transformer is expected to effectively capture spatial and temporal correlations between joints, facilitating action categorization without the rigid constraints imposed by GCNs on the human body's articulation. Despite this, there have been limited attempts to apply Transformer models to this domain[23], with some approaches combining GCNs with Transformers[24] or utilizing a multi-task learning framework[25].

The challenge of leveraging self-attention to learn meaningful spatial–temporal relationships between joints and extract representative motion features remains unresolved. Moreover, most existing Transformer-based approaches compute global one-to-one joint relationships in both spatial and temporal dimensions, which often overlook the significance of spatial interactions among key local joints and the short-term temporal dynamics critical for detecting action-relevant patterns. On one hand, since not all joints contribute equally to action recognition[1], these methods may suffer from the inclusion of irrelevant or noisy joints, with the self-attention mechanism amplifying their impact and potentially degrading recognition performance. On the other hand, given that the standard Transformer lacks inductive bias[26] for capturing the local structure of temporal data, these methods struggle to model effective temporal relations among joints when dealing with long input sequences.

Human skeletal data provides complementary information to RGB images and optical flow frames, offering a more structured representation of human actions. The human body can be viewed as a system composed of the trunk and limbs, where actions are defined by the movements of joints and bones. Skeletal motion can be represented as sequences of 2D positions of key human joints. With advancements in pose estimation networks, it is now possible to obtain highly accurate joint coordinates. These reliable skeletal data allow for the modeling of intra-person spatial–temporal dynamics, enabling the extraction of effective motion representations. In this article, we propose a multi-modal spatial–temporal fistfight detection network (MSTFDet) with two modules to capture both intra-person and inter-person relationships. For RGB image inputs, we introduce the Context-Aware Encoded Transformer, designed to model interactions between individuals and their surrounding environment, extracting informative Context-Aware features. In physical violence, individuals not only respond to their environment but also react to the behaviors of others in the fighting environment. To capture these interactions, we employ both a Spatial Transformer and a Temporal Transformer to infer spatial and temporal relationships between individuals. For skeletal inputs, we design Spatial GCN (S-GAN) and Temporal (T-GAN) to model intra-person spatial and temporal dynamics within the human body pose points.

Our main contributions are as follows:

- We propose MSTFDet, a multimodal spatio-temporal fistfight detection framework that combines RGB and skeletal data to enhance violence recognition in video sequences. The model leverages both visual and structural information, providing a comprehensive solution for detecting complex fistfight actions in dynamic environments.
- A new Context-Aware Encoded Transformer is introduced, which effectively captures the interaction between individuals and their surrounding environment. By utilizing contextual elements, the CAET enriches individual action features, enhancing the model's ability to accurately detect fistfight behaviors.
- The model integrates Spatial Transformer (STrans) and Temporal Transformer (TTrans) to capture spatio-temporal relationships between individuals. Additionally, it incorporates a novel combination of S-GCN and T-GCN for modeling intra-person dynamics, allowing for the effective detection of both individual and group actions.
- We conduct an ablation study to highlight the individual contributions of the RGB and BodyPose modules, showcasing the performance benefits of integrating both modalities. The analysis reveals the importance of incorporating spatial and temporal relationships for accurate fistfight detection in video data

The remainder of this paper is organized as follows: "Literature review" section reviews the related work, focusing on weakly labeled learning, federated learning, and efficient fine-tuning strategies for medical image

segmentation. "Methods" section details the proposed methodology, including the architectural design of Fed3D-KDNet, the federated aggregation strategy, and the hybrid knowledge distillation framework. "Experiment and result" section presents the experimental setup, datasets, evaluation metrics, and results obtained from both centralized and federated learning paradigms. "Discussion" section provides an in-depth discussion of the findings, highlighting the advantages and limitations of the proposed approach. Finally, "Conclusion" section concludes the paper by summarizing the key contributions, discussing potential applications, and outlining future research directions.

Literature review

Appearance-based violent detection

Appearance-based violence detection methods[27] extract visual features from RGB and optical flow data to model violent behaviors. Early approaches relied on 2D-CNN and 3D-CNN models[28] using optical flow as input to a 2D-CNN to capture motion features. Reference[29] extended 2D convolutions to 3D to improve performance, while Ref.[30] introduced FightNet, which computes optical flow to generate acceleration fields for representing violent interactions. Reference[31] utilized convolutional LSTM networks for violence detection.

Further advancements include a multi-stream CNN framework[32], which processes spatial, temporal, and spatiotemporal features. This approach efficiently learns environmental and motion patterns relevant to violent behaviors, including fistfights, through modified optical flow and motion energy images. However, appearance-based methods[33,34] struggle in complex scenes, with varying lighting and group behavior. Efforts to reduce the computational complexity of 3D-CNN models include lightweight CNN[35], which processes 16-frame sequences for efficient violence detection. Reference[36] proposed a DenseNet-based 3D-CNN to reduce parameters through feature reuse. Reference[37] introduced a method that uses 3D convolutions to model spatial–temporal relationships in violent interactions, demonstrating strong efficiency and real-world applicability. In contrast, unsupervised methods have also gained attention. Reference[38] proposed Double-AE, a double-stream AutoEncoder that filters background noise and extracts spatial–temporal features for violence detection. To address the scarcity of labeled violence data, Ref.[39] introduced an unsupervised Spatio Temporal Action Translation (STAT) network for detecting violent behaviors.

More recently, Efficient-X3D[40] applied transfer learning with pre-trained video models for anomaly detection in crowded scenes, improving computational efficiency and reducing overfitting. PiercingEye[41] proposed a dual-space learning framework combining Euclidean and hyperbolic feature spaces, achieving fine-grained discrimination of visually similar events using a hyperbolic vision-language guidance strategy. Light-WVAD[42], a lightweight weakly supervised framework, further reduced model complexity and label noise using adaptive instance selection and multi-level temporal correlation attention. Event-based data has also emerged as a promising direction; UCF-Crime-DVS[43] introduced the first event-driven anomaly detection dataset based on Dynamic Vision Sensors (DVS), enabling high-temporal-resolution representations for violence detection.

BodyPose-based violent detection

ST-LSTM[44] employs recurrent neural networks (RNNs) to model the temporal dynamics inherent in skeletal data. Specifically, the structural information of the human skeleton is input into a LSTM network, which captures the temporal evolution of joint positions across consecutive frames. This approach leverages the sequential nature of skeletal movements, enabling the model to learn the temporal dependencies between individual joint movements over time. The LSTM's ability to retain long-term dependencies in the data makes it particularly suitable for modeling complex, continuous actions, such as those involved in fistfights, where the coordination and timing of movements are crucial for accurate recognition. In contrast, convolutional neural network (CNN)-based approaches encode the temporal dynamics of skeletal data by converting skeleton joints into matrix representations, treating the sequences as image-like structures. These matrices, which encapsulate both spatial (joint locations) and temporal (movement across frames) information, are processed by CNNs to recognize actions in a manner akin to image classification tasks. This technique allows CNNs to exploit their ability to capture spatial hierarchies in data while learning temporal patterns. For instance, Hierarchical Co-occurrence Networks (HCN)[45] utilize an end-to-end learning framework where CNNs are employed to automatically extract hierarchical co-occurrence features from skeleton sequences. These features capture both the spatial configuration of joints and their temporal transitions, enabling robust action recognition in dynamic settings.

ST-GCN[46] represents a significant advancement in the field by extending the concept of neural networks to graph structures. ST-GCN is designed to process skeletal data by treating human skeleton joints as nodes in a topological graph, with the edges representing the relationships between joints. This spatiotemporal graph model allows for a more natural and flexible representation of human motion, as it directly captures the spatial and temporal dependencies between joints. The GCN used in ST-GCN acts as the backbone for behavior recognition, applying convolutions over the graph structure to learn both the spatial configuration of joints and their temporal evolution. However, while ST-GCN is effective in modeling skeleton-based actions, it faces challenges when integrating with other modalities, such as RGB data and optical flow, as its complexity increases linearly with the number of individuals in the scene. This scalability issue can limit its applicability in real-time or large-scale action detection tasks.

Recent approaches have sought to address these limitations by treating skeleton sequences as 3D point clouds, which allows for the learning of contextual relationships between individuals from skeletal points. This is particularly relevant for the detection of violent behaviors, such as fistfights, where multiple individuals are often involved in complex, highly dynamic interactions[1,47]. In this context, skeleton-based methods offer a structured and interpretable representation of human actions, but they often struggle to capture the fine-grained local motion patterns that are critical for recognizing violent interactions in real-world scenarios.

To enhance contextual modeling, Su et al.[48] introduced a semantic-driven dual consistency learning framework that aligns visual and semantic features under weak supervision. EventVAD[49] also contributes to this direction by proposing a training-free method that segments events using graph-based temporal boundaries, eliminating the need for manual annotations. Furthermore, Fed-WS-VAD[50] explored the federated learning paradigm for weakly supervised anomaly detection across heterogeneous scenes without sharing raw video data, ensuring privacy preservation and robustness to domain shifts.

To address these challenges, our proposed method enhances the representation of skeletal key points by fusing motion and appearance information across consecutive frames. This fusion allows the model to capture both the local motion patterns of keyframes and the broader spatial context, thereby improving its ability to detect violent fistfights. By integrating skeletal and appearance data within a unified framework, we are able to leverage the complementary strengths of each modality, leading to more accurate and robust detection of violent behaviors in video sequences.

Methods

The proposed network integrates two primary components: the RGB Image module and the BodyPose-Points Data module, both of which work together to capture comprehensive spatio-temporal features critical for detecting fistfight actions in video data, as shown in Fig. 1. The RGB Image module processes raw video frames to capture inter-person interactions and dynamic spatial–temporal contexts. Using a backbone network like EfficientNet, high-level spatial features are extracted from each frame, refined through RoIAlign to ensure precise focus on individual actions within the scene. The Context-Aware Encoded Transformer (CAET) enhances this by modeling the interactions between individuals and their environment, leveraging attention mechanisms to encode critical relational features that improve understanding physical confrontations. To further enhance this module, a Spatial Transformer (STrans) captures the spatial relationships between individuals, treating each person as a node in a graph. At the same time, a Temporal Transformer (TTrans) tracks the evolution of these relationships over time. This combined spatial–temporal analysis provides a robust representation of the dynamic interactions during violent altercations. Meanwhile, the BodyPose-Points Data module focuses on human skeletal data, providing a structured representation of human actions through key joint positions extracted using a pose estimation model such as HRNet. This skeletal data offers critical intra-person dynamics by modeling how joints move about one another.

The Spatial–Temporal GCN (ST-GCN) is employed to model these intra-person dynamics, treating joints as nodes in a graph and capturing spatial relationships within a single frame, as well as tracking joint movements over time. The spatial component of ST-GCN propagates information between connected joints, enabling the model to detect specific movement patterns such as punches or grappling motions, while the temporal component captures the progression of these movements across frames. The final step involves fusing the outputs from both the RGB Image and BodyPose-Points Data modules. The extracted features are concatenated, creating a unified representation that captures both the visual context and the structural dynamics of fistfight

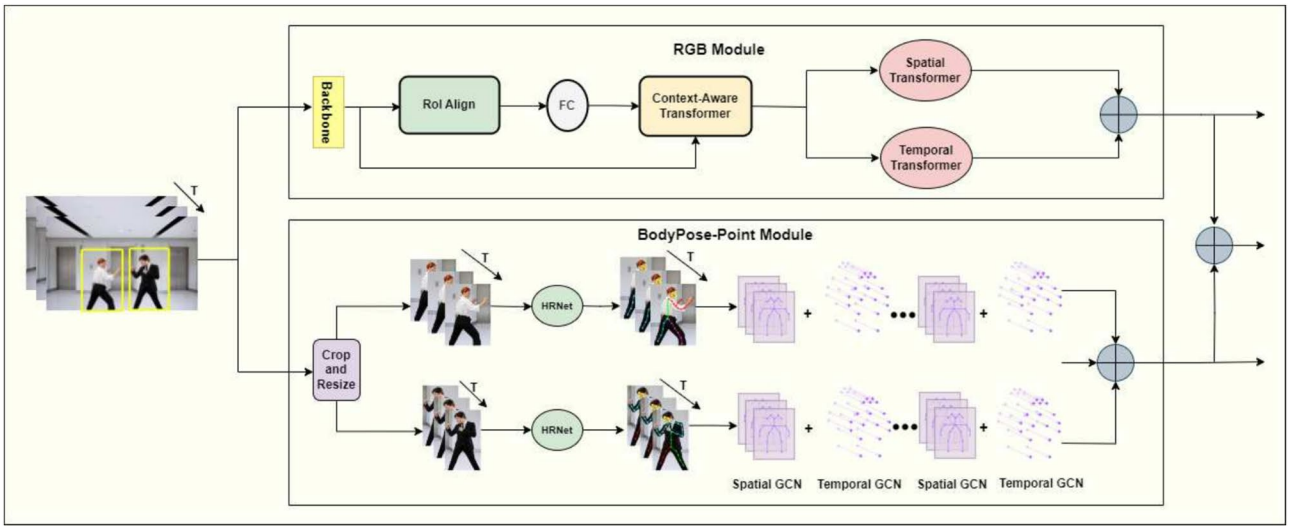

Fig. 1. The architecture of our proposed Multimodal Spatio-Temporal Fistfight Detection model (MSTFDet). The model consists of two main modules: the RGB Module and the BodyPose-Point Module. The RGB Module utilizes RoI Align to extract spatial features from RGB frames, followed by a Context-Aware Transformer, which encodes the relational features between individuals and their environment. The spatial and temporal relationships are modeled using Spatial Transformer and Temporal Transformer, respectively. The BodyPose-Point Module processes skeleton-based data using HRNet to extract keypoint features. These features are further processed through Spatial GCN and Temporal GCN layers to capture intra-personal and inter-personal dynamics. The outputs from both modules are fused for final action detection, leveraging both RGB and pose information for enhanced performance in fistfight detection.

actions. This multimodal fusion enhances the model's ability to detect complex and subtle behaviors by utilizing complementary information from both visual and skeletal data. The fused features are then passed through classifiers that predict the class.

RGB modality

In the initial step, T-frame images are used as input to extract individual features through the Image Feature Extractor. Subsequently, the Context-Aware Encoded Transformer (CAET) integrates contextual information to refine and enhance these individual features. Given the dynamic nature and interactions inherent in both individual fistfight actions and group activities, the Spatial Transformer (STrans) and Temporal Transformer (TTrans) is employed to explore the spatio-temporal relationships within the enhanced individual features. Finally, the outputs of the STrans and TTrans are concatenated, followed by the application of a pooling layer to derive the group-level features from the Image module. Unlike STrans, which models spatial relations across different individuals, S-GCN focuses exclusively on intra-person spatial configurations, leveraging skeletal topologies to capture joint-level anatomical constraints.

Although our system processes a fixed number of frames T centered around the key action frame, we mitigate the limitations associated with this temporal window size through several architectural design choices. First, the use of the Temporal Transformer (TTrans) enables the model to capture long-range dependencies across frames within the T-length window using self-attention, without relying on sequential operations that often dilute memory of earlier frames. Furthermore, the sliding-window processing strategy ensures that overlapping windows cover the entire video, allowing delayed or prolonged violent actions to be captured in adjacent sequences. Thus, while the temporal length T is fixed, our system effectively maintains robustness to long-range or delayed behaviors through attention-based modeling and temporal window redundancy.

RGB feature extractor

The RGB module processes the input video by receiving T-frame sequences, where each sequence is centered around the key frame that contains the labeled action. This method ensures that both the preceding and following temporal contexts are captured effectively, aiding in the recognition of dynamic actions in violent scenarios. To extract meaningful spatial–temporal features from these input frames, we employ a robust backbone network, such as EfficientNet or ResNet, which is specifically designed for high-performance feature extraction in vision tasks. This backbone processes the T-frame sequences to generate feature maps, denoted as $X \in \mathbb{R}^{T\times D\times H\times W}$, where T represents the temporal dimension, D is the depth of the feature map, and H and W correspond to the spatial height and width of each frame, respectively. To isolate the actions of individual persons within the scene, we apply Region of Interest (RoI) alignment using the bounding boxes of N individuals. RoIAlign, as described in Ref.[51], is particularly advantageous because it accurately aligns the features of each individual, maintaining the spatial integrity of their movements and positions, which is crucial for fine-grained action detection such as fistfights. The RoIAlign operation crops and resizes the feature maps to focus on the regions within the bounding boxes, ensuring that each individual is represented by distinct spatial features.

After RoIAlign, the refined individual feature maps are passed through a fully connected (FC) layer, transforming them into d-dimensional vectors. These vectors represent the features of each individual in the scene across the T frames, and are expressed as $X_I \in \mathbb{R}^{T\times N\times d}$, where N represents the number of individuals, and d is the feature dimension of each individual. The FC layer serves to condense the high-dimensional feature maps into a more compact and discriminative representation that highlights the key attributes of each person's actions, such as motion patterns, posture, and contextual interactions with others. By converting the extracted feature maps into d-dimensional representations, this process enables the system to effectively capture critical spatial and temporal details associated with violent interactions, such as the speed, direction, and intensity of movements, which are essential for accurately identifying fistfights. The resulting X_I matrix contains rich spatial-temporal information, providing a robust foundation for the subsequent stages of action detection, where further refinement and classification of violent behaviors, including punching, kicking, and grappling, can occur within the broader spatial context of the scene.

Bounding-box generation

Accurate localisation of each participant is a prerequisite for all subsequent stages of *MSTFDet*, because the same bounding boxes drive (i) RoIAlign for the RGB branch, (ii) cropping of input patches for HRNet in the BodyPose branch, and (iii) the positional encodings consumed by both CAET and STrans. We therefore designed a dedicated, lightweight pipeline that is *tightly coupled* to the computational budget and latency constraints of the full model. We start with the anchor-free YOLOv8-x detector (640 px input) and fine-tune it for 20 epochs on an in-house union of CrowdHuman, COCO-Person, and the training splits of both SCFD and RWF-2000. This domain-specific fine-tuning raises recall for small, partially occluded figures by $+6.1\,\%\ \mathrm{AP}_{50}$ on the SCFD validation set, while leaving inference cost at 7.4 ms/frame on an RTX 4090.

Standard Greedy-NMS fails whenever inter-person IoU exceeds 0.6, a frequent situation during grappling or shoving. We therefore adopt *Soft-NMS* with linear decay and $\tau_{\mathrm{IoU}} = 0.45$. Additionally, each candidate box is associated with a visibility score:

$$v = \frac{\sum_{(x,y)\in B} 1\{\sigma_{xy} \geq 0.3\}}{|B|}, \tag{1}$$

where σ_{xy} is the detector's objectness heat-map. Boxes with $v < 0.25$ are discarded, preventing spuriously large boxes that arise under heavy occlusion from contaminating RoIAlign.

Transient full occlusion (1–3 frames) still breaks temporal consistency and degrades TTrans. We bridge these gaps with a lightweight *DeepSORT* formulation in which the Kalman state vector is $(x, y, w, h, \dot{x}, \dot{y}, \dot{w}, \dot{h})$ and the appearance descriptor is the 256-D penultimate-layer embedding of YOLOv8-x. A track is accepted if its mean detection confidence $\bar{c} \geq 0.25$ and its age ≤ 30 frames; otherwise it is flushed to avoid drift. When the instantaneous density exceeds 3.5 persons/m^2 (estimated by kernel density on box centres), we enable motion-guided propagation: each box from frame $t-1$ is warped to frame t using RAFT-Small optical flow, and the resulting hypotheses are fused with the detector outputs via *Weighted Box Fusion*. This recovers up to $+12.4\,\%$ missed true positives at an amortised overhead of 1.3 ms/frame.

Context-aware encoded transformer

The contextual feature representation X comprises both visual and positional components that are distributed across $H \times W$ spatial elements. Each of these elements contributes differently to the final decision-making process for a given target individual, depending on the significance of the surrounding environment and its interaction with the individual in question. The attention mechanism within the transformer architecture is designed to account for these varying contributions by dynamically assigning adaptive weights to each element. These weights are computed based on the degree of relevance each contextual element holds in relation to the target individual's action, effectively capturing complex interdependencies within the scene. To facilitate the integration of individual features X_I with the contextual feature maps X, both are projected into a shared d_E-dimensional subspace. This transformation yields $X'_I \in \mathbb{R}^{T \times N \times d_E}$ and $X' \in \mathbb{R}^{T \times HW \times d_E}$, where N represents the number of individuals and HW corresponds to the spatial dimensions of the contextual map. The shared subspace ensures that both the individual-specific features and the environmental context are represented in a common framework, enabling efficient interaction between them. In this projection, the transformed individual features X'_I serve as the query Q, while the contextual features X' act as the key K in the attention mechanism. The attention mechanism calculates the adaptive weights by determining the similarity between Q and K, leading to a weighted sum V that represents the enriched features containing both individual and contextual information. Archtecture of CAET is shown in Fig. 2.

The attention mechanism in CAET computes adaptive weights through a scaled dot-product operation between queries (individual features) and keys (contextual features). Formally, given the projected queries $Q = X'_I \in \mathbb{R}^{T \times N \times d_E}$, keys $K = X' \in \mathbb{R}^{T \times HW \times d_E}$, and values $V = X'$, the attention weights are calculated as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_E}}\right) V, \tag{2}$$

where the scaling factor $1/\sqrt{d_E}$ prevents gradient vanishing issues that can occur with large values of d_E. The softmax operation normalizes the attention scores across the spatial dimension *HW*, ensuring that the weights sum to 1 for each query position. For multi-head attention with *h* heads, the queries, keys, and values are first linearly projected into *h* different subspaces:

$$Q_i = QW_i^Q, \quad K_i = KW_i^K, \quad V_i = VW_i^V, \tag{3}$$

where $W_i^Q, W_i^K, W_i^V \in \mathbb{R}^{d_E \times d_E/h}$ are learnable projection matrices for head *i*. The attention outputs from all heads are concatenated and projected:

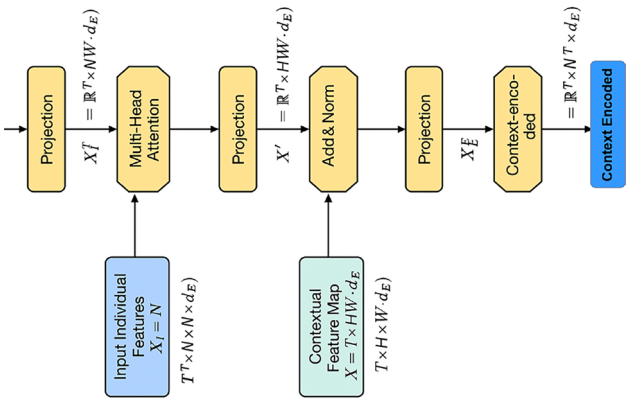

Fig. 2. Architecture of the CAET module. The module takes as input the individual features and the contextual feature map extracted from the RGB frames. Both are projected into a shared subspace and passed through a multi-head attention mechanism. The resulting features are normalized, further transformed through projection layers, and fused to produce the context-encoded individual feature representation $X_E \in \mathbb{R}^{T \times N \times d_E}$, capturing inter-person and environment-aware interactions essential for fistfight detection.

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, ..., \text{head}_h)W^O, \tag{4}$$

where $W^O \in \mathbb{R}^{d_E \times d_E}$ is a learnable output projection matrix. The positional encodings are fused with the visual features through element-wise addition after the projection into the shared subspace:

$$X'_{I,pos} = X'I + PE_I, \quad X'pos = X' + PE, \tag{5}$$

where $PE_I \in \mathbb{R}^{T \times N \times d_E}$ and $PE \in \mathbb{R}^{T \times HW \times d_E}$ are the positional encodings for individuals and contextual features respectively, computed as described in Eqs. (1)–(4).

Since transformers inherently lack the inductive biases introduced by recurrence and convolution, which are typically beneficial for capturing local spatial and temporal correlations, positional encodings are introduced to compensate for this limitation[19]. These positional encodings are essential for providing the model with information about the relative positions of elements within the spatial and temporal dimensions. Specifically, the positional encoding PE for each element is computed using sine and cosine functions of different wavelengths, ensuring that each position is uniquely encoded in a way that the model can interpret. For each individual feature, the center coordinates (x_i, y_i) of the bounding box are encoded to inject positional information, where x_i and y_i represent the horizontal and vertical center positions of the bounding box, respectively. These coordinates are encoded as follows:

$$PE_{(\text{pos, 2k})} = \sin\left(\frac{\text{pos}}{10000^{2k/L}}\right), \tag{6}$$

$$PE_{(\text{pos, 2k+1})} = \cos\left(\frac{\text{pos}}{10000^{2k/L}}\right). \tag{7}$$

For each individual's feature, the center coordinates (x_i, y_i) of their bounding box are encoded as:

$$\text{pos} = \begin{cases} x_i, & k \in \{0, 1, \ldots, L/4 - 1\} \\ y_i, & k \in \{L/4, L/4 + 1, \ldots, L/2 - 1\}. \end{cases} \tag{8}$$

This encoding ensures that positional information about the bounding box is embedded in the features, providing the model with spatial context regarding the location of the individual within the scene. Similarly, for the contextual feature map X', the spatial coordinates (x, y) of each element are also encoded using the same positional encoding scheme. These encodings provide the model with the necessary spatial awareness, allowing it to interpret the relationships between different elements within the context as:

$$\text{pos} = \begin{cases} x, & k \in \{0, 1, \ldots, L/4 - 1\} \\ y, & k \in \{L/4, L/4 + 1, \ldots, L/2 - 1\}. \end{cases} \tag{9}$$

The shared d_E-dimensional subspace plays a crucial role in aligning the semantic representations of individual-specific features X_I and contextual features X. By projecting both X_I and X into the same embedding space, the model ensures that their representations become directly comparable and mutually informative. This alignment facilitates richer cross-attention, as it allows the transformer to compute similarity scores and perform interactions between individual queries and contextual keys/values in a semantically consistent manner. The shared subspace acts as a unifying framework that enhances the granularity and interpretability of interactions, enabling the model to focus on contextually relevant cues (e.g., nearby threats, sudden movements) that influence or are influenced by the target individual's actions. Importantly, this also allows the attention mechanism to suppress irrelevant background noise by favoring contextual elements that correlate with the individual's behavior within the scene.

Finally, the result of these operations is the context-encoded individual feature representation $X_E \in \mathbb{R}^{T \times N \times d_E}$, which integrates critical environmental information into the individual feature space. By encoding the interactions between the individual and their surrounding context, this enriched feature representation improves the model's ability to detect and understand complex actions, such as fistfights, within dynamic and cluttered environments. The incorporation of positional encodings further enhances the model's spatial–temporal reasoning.

We explicitly fuse visual and positional components at the representation level prior to input into the transformer-based modules. After extracting individual-level appearance features from the RGB frames using RoIAlign and projecting them into a compact feature space, we enrich each individual's representation by concatenating it with a positional embedding derived from the center coordinates (x_i, y_i) of the associated bounding box. These positional encodings are computed using sinusoidal functions, as formulated in Eqs. (1)–(3), enabling the model to retain spatial locality information in a format that is both continuous and differentiable. The resulting enriched token, which now encapsulates both appearance and position, is projected into a shared subspace and subsequently serves as the input to all transformer-based modules: the CAET, Spatial, and Temporal Transformer. This fused representation ensures that the self-attention mechanisms can jointly reason over semantic content and spatial configuration.

Spatial transformer

The spatial Transformer (STrans) plays a critical role in modeling the spatial interactions between individuals by treating each person as a node within a graph structure, effectively capturing the spatial relationships within the context of a fistfight as shown in Fig. 3. By interpreting individuals as graph nodes, the STrans module aggregates

information from neighboring nodes (individuals), thereby enhancing the feature representation for each individual through the integration of surrounding contextual cues. This approach allows for the spatial-context of inter-person interactions to be encoded, enabling the model to grasp the dynamic interplay of individuals engaged in physical altercations. One of the key mechanisms of the STrans is its ability to assign varying levels of importance to different neighboring individuals based on their relative proximity and the nature of their interactions. This is achieved through an attention mechanism that weighs the contribution of each neighboring node to the target node's feature representation, effectively highlighting the most relevant spatial relationships. The adaptive attention mechanism ensures that individuals who are spatially closer and more involved in the action are given more weight, which enhances the model's ability to accurately represent the inter-person dynamics within a scene of physical violence. It is important to note that the STrans module captures inter-person spatial interactions within the RGB modality, and is fundamentally distinct from the intra-person spatial modeling conducted by the S-GCN in the skeletal stream.

The STrans module processes the temporally structured spatial-context encoded individual features X_E by treating the temporal dimension as the batch dimension, thus allowing for simultaneous processing of multiple time frames. The module leverages projection functions to map the spatial-context encoded features into query (Q_S), key (K_S), and value (V_S) representations, which are integral components of the self-attention mechanism. Through the computation of attention weights between individuals (nodes), the STrans captures critical spatial interactions that influence the overall detection of fistfights. In dynamic surveillance scenarios where the number of individuals and their spatial positions change over time, we construct a new graph for each frame based on the current set of detected individuals. Each node represents a person, and the edge weights between nodes are computed based on a hybrid metric that combines both spatial proximity and motion similarity. Spatial proximity is measured using the Euclidean distance between bounding box centers, while motion similarity is derived from the cosine similarity between estimated velocity vectors of individuals (computed via optical flow or bounding box displacement).

Formally, the edge weight w_{ij}^t between individuals i and j at frame t is defined as:

$$w_{ij}^t = \exp\left(-\alpha \cdot \|p_i^t - p_j^t\|_2^2\right) \cdot \left(1 + \beta \cdot \mathrm{sim}(v_i^t, v_j^t)\right), \tag{10}$$

where p_i^t and p_j^t represent the 2D spatial positions of individuals i and j, v_i^t and v_j^t denote their estimated motion vectors, and $\mathrm{sim}(\cdot,\cdot)$ is the cosine similarity function. The hyperparameters α and β control the influence of spatial and motion-based affinity, respectively. Graph topology is updated at each frame, and edges with weights below a threshold τ are pruned to retain only significant interactions. To ensure temporal stability in the presence of transient occlusions or noisy detections, we optionally apply an exponential moving average over past edge weights.

The output of the attention mechanism X_S incorporates the spatial context of the scene, enhancing the feature representation with inter-person relationships that are crucial for detecting and understanding violent interactions. To further enrich the spatial representation, positional encoding is applied to retain the spatial structure of the data, ensuring that the model can distinguish between individuals based on their spatial arrangement within the scene. The positional encoding mechanism follows the same method described in the CAET, where both the visual and positional components of each individual's features are encoded to preserve their spatial location relative to the environment and other individuals. This ensures that the spatial relationships are maintained throughout the network, allowing the STrans to effectively model the spatial dynamics in the context of fistfight detection.

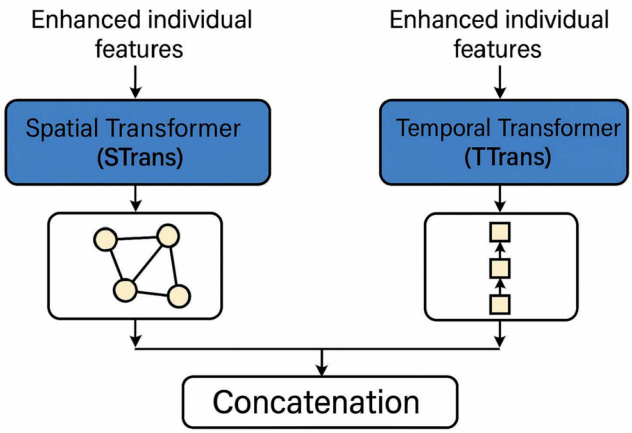

Fig. 3. Architecture of the Spatial Transformer (STrans) and Temporal Transformer (TTrans) modules within the RGB stream. The STrans captures inter-person spatial interactions in each frame by modeling attention-based relationships among individuals. The TTrans captures the evolution of each individual's actions across frames, focusing on inter-frame dependencies. Together, they enable comprehensive modeling of group-level spatio-temporal dynamics in RGB video.

In our implementation of the STrans, the temporal dimension is treated as a batch dimension during self-attention computation. This design choice enables the model to process the spatial relationships independently for each frame, thus simplifying the computational graph and enabling frame-wise inter-person reasoning. However, this implies that explicit temporal dependencies are not modeled within the STrans module itself. To mitigate the risk of losing important temporal context, TTrans is explicitly designed to capture inter-frame dependencies after spatial reasoning has been applied. TTrans operates across the temporal axis of encoded individual representations and employs self-attention mechanisms to model both short- and long-term temporal patterns. This sequential decoupling first modeling spatial dependencies per frame (via STrans), followed by temporal dependencies across frames (via TTrans) allows the system to maintain full temporal reasoning capacity. Furthermore, the fused multimodal architecture, including temporal GCNs in the bodypose stream, reinforces temporal continuity by modeling joint-wise dynamics across consecutive frames.

Temporal transformer

Temporal dynamics play a crucial role in understanding the progression of individual actions, particularly in scenarios involving group activities like fistfights, where interactions unfold over time. To effectively capture these dynamics, we introduce the Temporal Transformer (TTrans), which is designed to encode the temporal evolution of individual features. The TTrans is integral to modeling the sequence of actions over consecutive frames, ensuring that the temporal relationships between individuals are preserved and leveraged for improved action recognition as shown in Fig. 3. Previous approaches, such as the Inflated 3D ConvNet (I3D) models, attempted to capture temporal information by expanding 2D convolutional networks into the temporal domain to process image sequences directly[52]. However, these models often neglected the fine-grained temporal dynamics of individual actions, instead focusing on aggregated or pooled representations across the entire sequence. This pooling operation[53] can result in a loss of critical temporal information that is essential for distinguishing nuanced actions, particularly in the context of complex, fast-paced interactions like fistfights.

In contrast, the TTrans leverages an attention-based mechanism that is well-suited for capturing both short-term and long-term dependencies across frames. The self-attention mechanism within the TTrans allows it to adapt to the varying importance of different time steps, ensuring that the features of individuals are refined based on their temporal context. This attention mechanism not only identifies relevant temporal patterns but also mitigates the risk of losing important individual-level temporal information, which is a common drawback of traditional pooling techniques. The TTrans operates by treating the spatial dimension of the encoded individual features X_E as the batch dimension, allowing it to process the temporal relationships between different frames. To ensure that the temporal sequence is properly aligned with the model's architecture, we perform a dimensional swap on X_E, transforming it into $X'_E \in \mathbb{R}^{N \times T \times d_E}$. This reshaping ensures that the temporal dimension becomes the primary focus of the self-attention mechanism, allowing the model to capture the evolution of individual actions over time. To further enhance the temporal encoding, positional encoding is applied within the TTrans, ensuring that each time step t is encoded with respect to its position in the sequence. This approach follows a sinusoidal positional encoding scheme, where the position of each frame is encoded into a fixed-dimensional representation, ensuring that the model retains a sense of temporal order. The positional encoding is defined as:

$$\text{pos} = t, \quad t \in [0, T]. \tag{11}$$

Bodypose-points modality

We extract human bodypose-points data from RGB images using a bodypose-points Data Extractor. ST-GCNs are employed to capture intra-person spatio-temporal dynamics. These features are then pooled together to generate the group-level features in the bodypose-points module.

Bodypose-points feature extractor

The bodypose-points data encapsulates intra-person interactions, differing from the inter-person interactions modeled in the Image module. These interactions are represented as sequences of 2D coordinates corresponding to human joints. We utilize HRNet[16], which is pre-trained on the COCO dataset, as a pose estimation network. Based on the bounding boxes of each individual, the RGB images are cropped and resized to 192×256.

Spatial GCN

The spatial relationships between human joints, critical for accurately modeling human motion, are naturally represented as a graph structure due to the irregular and dynamic nature of human skeletal data. In our framework, we employ a Spatial GCN (S-GCN) to model intra-person spatial relations, which are essential for understanding the interactions between different joints within a single person as shown in Fig. 4. Specifically, we define a spatial graph $G_t = (V_t, E_t)$ for the human joints in each frame t, where the set of nodes $V_t = v_{ti} | i = 1, \ldots, N_{\text{joint}}$ represents individual joint positions. Each node v_{ti} corresponds to a specific joint, with attributes such as its spatial coordinates and associated confidence scores derived from pose estimation networks The edges $E_t = e_{ij} | i, j = 1, \ldots, N_{\text{joint}}$ represent the pairwise relationships between joints, capturing the interactions between connected body parts. These edges reflect the physical and anatomical relationships between joints, such as the connection between the elbow and shoulder or the knee and hip. To model these connections mathematically, we construct an adjacency matrix A^S, which serves as a fundamental component of the GCN. The adjacency matrix encodes the connectivity between joints, specifying which joints are directly connected based on anatomical constraints.

In modeling joint interactions, it is necessary to impose certain constraints to reduce complexity while maintaining the integrity of the human body's movement patterns. We assume that joints only interact with other

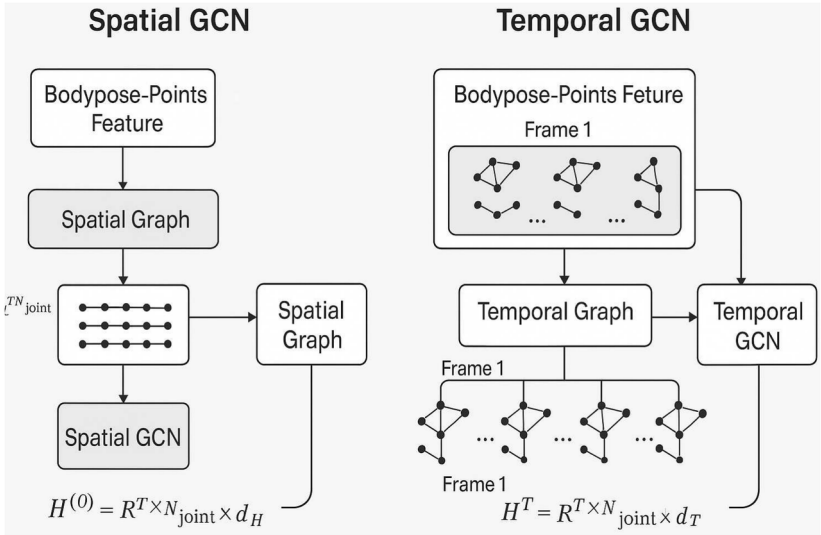

Fig. 4. Detailed architecture of the Spatial GCN (S-GCN) and Temporal GCN (T-GCN) used in the bodypose-points stream. The S-GCN constructs a joint-level graph per frame to model intra-person anatomical relationships based on skeletal topology. The T-GCN connects identical joints across consecutive frames to capture the temporal evolution of motion. Each node corresponds to a human joint, and edge connections represent physical or temporal dependencies.

joints within a certain range, reflecting the physical limitations of joint movements. Therefore, the neighborhood of a joint node v_{ti} is defined as $N(v_{ti}) = v_{tj}|d(v_{ti}, v_{tj}) \leq 1, j \neq i$, where $d(v_{ti}, v_{tj})$ is the distance between joints *i* and *j*. We assume that joints only interact with other joints within a certain range, reflecting the physical limitations of joint movements. Specifically, the distance threshold $d(v_{ti}, v_{tj}) \leq 1$ used to define the local neighborhood $N(v_{ti})$ refers to the *topological distance* on the human skeletal graph, rather than the Euclidean distance in image space. This means that a joint is considered a neighbor if it is directly connected or adjacent in the anatomical skeletal structure, as defined by standard kinematic tree connections (e.g., shoulder to elbow, elbow to wrist). This design aligns with physiological constraints and preserves the semantic structure of the human body. With topological adjacency, the model focuses on meaningful anatomical dependencies rather than arbitrary proximity in 2D or 3D coordinate space, which could be distorted due to camera perspective or projection artifacts.

This localized neighborhood approach enables the model to focus on meaningful interactions between anatomically connected joints while avoiding unnecessary computational complexity arising from distant joints that do not directly influence each other. To further refine the modeling of joint interactions, we consider the physiological basis of human movement. Joint motions are driven by muscle activity, which can be categorized into two primary types: shortening, where muscles contract to pull weight toward the body, and lengthening, where muscles extend to push weight away from the body. These two types of muscle-driven motion result in different spatial patterns of joint movement, which we represent by dividing the neighborhood $N(v_{ti})$ into two distinct sets: proximal and distal. The proximal set contains joints that are closer to the center of the body, reflecting movements that involve pulling weight toward the torso. Formally, this set is defined as:

$$\text{Proximal} = \{v_{tj}|d(v_{tj}, v_{\text{center}}) < d(v_{ti}, v_{\text{center}}), v_{tj} \in N(v_{ti})\}, \tag{12}$$

$$\text{Distal} = \{v_{tj}|d(v_{tj}, v_{\text{center}}) > d(v_{ti}, v_{\text{center}}), v_{tj} \in N(v_{ti})\}. \tag{13}$$

Here, v_{center} refers to the neck joint, which is treated as the center of the body. The adjacency matrix A^S_{ij} depends on the set to which the neighboring joint v_{tj} belongs in the tth frame. The joint nodes are stacked to form $H^{(0)}$.

Temporal GCN

The Temporal GCN (T-GCN) is employed to model the temporal dynamics of human joint movements across frames, which is crucial for capturing the evolution of actions over time as presented in Fig. 4. In the context of human action recognition, especially for complex activities like fistfights, temporal coherence is as important as spatial relationships. The T-GCN allows for a structured representation of these temporal dynamics by constructing a temporal graph $G_i = (V_i, E_i)$ for each joint *i*, where the set of nodes $V_i = v_{ti}|t = 1, 2, \ldots, T$ represents the state of joint *i* at different time frames *t*. This construction enables the model to analyze how each joint evolves over the course of an action, tracking its motion through consecutive frames. The edges $E_i = e_{tt'}|t, t' = 1, 2, \ldots, T, t' \neq t$ in the temporal graph represent the connections between a joint's positions in different frames.

Initialize: RGB Feature Extractor Θ_{RGB}, RoIAlign, CAET, STrans, TTrans
Initialize Pose Feature Extractor (HRNet) Θ_{Pose}, S-GCN, T-GCN
Initialize fusion and classifier layers
Set maximum temporal sequence length T, maximum individuals per frame N
for each video sequence V_i **do**
  Extract frames $F_{1:T}$, bounding boxes $B_{1:T}^{1:N}$
  for $t = 1, 2, \ldots, T$ **do**
    RGB Module:
    Extract frame-level features: $X_t \leftarrow \Theta_{RGB}(F_t)$
    Apply RoIAlign: $X_{RoI}^t \leftarrow RoIAlign(X_t, B_t^{1:N})$
    Compute individual features: $X_I^t \leftarrow FC(X_{RoI}^t)$
    Context-aware features: $X_E^t \leftarrow CAET(Q, K, V; X_I^t)$
    Spatial features: $X_S^t \leftarrow STrans(X_E^t)$
    Temporal features: $X_T^t \leftarrow TTrans(X_E^t)$
    Pool spatial-temporal RGB features: $X_{RGB}^t \leftarrow Pool([X_S^t, X_T^t])$
    BodyPose Module:
    Extract pose-points: $P_t^{1:N} \leftarrow \Theta_{Pose}(F_t, B_t^{1:N})$
    Spatial graph: $H_S^t \leftarrow S\text{-}GCN(P_t^{1:N})$
    Temporal graph: $H_T^t \leftarrow T\text{-}GCN(P_t^{1:N})$
    Pool pose features: $X_{Pose}^t \leftarrow Pool([H_S^t, H_T^t])$
    Fusion and Prediction:
    Fuse multimodal features: $X_{fused}^t \leftarrow [X_{RGB}^t, X_{Pose}^t]$
    Predict classes: $\hat{y}_t \leftarrow Classifier(X_{fused}^t)$
  end for
  Compute losses: $\mathscr{L}_1(y^G, \hat{y}^G)$ and $\mathscr{L}_2(y^I, \hat{y}^I)$
  Total loss: $\mathscr{L} \leftarrow \mathscr{L}_1 + \lambda \mathscr{L}_2$
  Optimize model via backpropagation (Adam optimizer)
end for

Algorithm 1. Pseudo-code of MSTFDet for multimodal spatio-temporal fistfight detection.

To efficiently capture the temporal context without overburdening the model with unnecessary data, we limit the analysis to a sliding window of M frames centered around the current frame t. This window-based approach ensures that the model focuses on the most relevant temporal information while avoiding the computational complexity of processing the entire video sequence. The adjacency matrix $A_{tt'}^T$, which governs the connections between different time frames, is defined as

$$A_{tt'}^T = \begin{cases} 1, & t - \frac{M-1}{2} \leq t' \leq t + \frac{M-1}{2} \\ 0, & \text{otherwise.} \end{cases} \tag{14}$$

The joint nodes v_{ti} in each frame are stacked to form $H^{(0)}$. These edges are critical for modeling the continuity of joint movement, linking the same joint across frames to capture its temporal evolution. This setup allows the model to understand the progression of joint motion, which is key to detecting dynamic actions like punches or grapples in fistfights. By encoding these temporal dependencies, the T-GCN ensures that the model does not treat each frame in isolation but rather as part of a continuous sequence that reflects real-world movement.

Multimodal feature fusion strategy

To effectively integrate the pose-based features extracted from HRNet with RGB-based spatial features processed through CAET, STrans, and TTrans, we employ a late fusion strategy after modality-specific spatiotemporal encoding. Specifically, for each frame t, the RGB module outputs a group-level spatiotemporal feature vector $X_{\text{RGB}}^t \in \mathbb{R}^d$, while the BodyPose module produces a corresponding skeletal feature vector $X_{\text{Pose}}^t \in \mathbb{R}^d$. These features are concatenated to obtain a unified multimodal representation:

$$X_{\text{fused}}^t = \left[X_{\text{RGB}}^t \,\|\, X_{\text{Pose}}^t\right] \in \mathbb{R}^{2d}. \tag{15}$$

This fused representation is passed through a feedforward fusion layer with a residual connection and layer normalization, producing the final spatiotemporal representation used for classification. This design preserves

the modality-specific information while allowing effective cross-modal interaction during the final prediction stage. Importantly, the temporal alignment is maintained by synchronizing the frame indices from which the RGB and pose features are extracted, ensuring one-to-one correspondence across modalities.

Loss function
To train the proposed end-to-end model for fistfight detection, we employ a cross-entropy loss function. The total loss is calculated as:

$$\mathscr{L} = \mathscr{L}_1(y^G, \hat{y}^G) + \lambda \mathscr{L}_2(y^I, \hat{y}^I), \tag{16}$$

where $\mathscr{L}_1$ and $\mathscr{L}_2$ represent the cross-entropy losses. Here, y^G and $\hat{y}^G$ denote the ground truth and predicted values of the group activity, while y^I and $\hat{y}^I$ represent the ground truth and predicted values of the individual action, respectively. The parameter λ is a hyperparameter used to balance the contributions of the two losses. Pseudo-code of our proposed MSTFDet model is outlined in Algorithm 1.

Experiment and result

Dataset

Surveillance camera fight dataset
We have utilized the Surveillance Camera Fight Dataset (SCFD)[54], which contains a comprehensive collection of real-world surveillance footage. This dataset is composed of 1000 video clips, each lasting between 5 and 15 s, capturing violent interactions in public settings such as streets, transportation hubs, and public institutions. The videos are recorded in real-time under varying lighting and environmental conditions, closely resembling scenarios that are commonly encountered in security surveillance applications. From this dataset, we have chosen a subset of specific action classes relevant to fistfight detection.

The action classes in the SCFD dataset were annotated manually by a team of trained human annotators. Each annotator was provided with the video sequences and a predefined list of violent behavior types, including punching, slapping, elbow strikes, choking, and grappling. Annotation was performed using frame-by-frame analysis in an offline tool, with each action labeled based on visual evidence of contact, intent, and body movement patterns. Multiple annotators cross-validated each instance to ensure labeling consistency and reduce subjectivity. In ambiguous cases, consensus was reached through majority agreement under expert supervision.

These classes include punching (150 instances), slapping (140 instances), elbow Strikes (150 instances), choking (155 instances), and grappling (120 instances). These actions represent typical forms of aggressive physical behavior observed during fistfights, involving dynamic, close-quarters interactions between individuals. Punching and Slapping are the most frequent violent actions in the dataset, while Elbow Strikes and Choking contribute a unique dimension of upper and lower body engagement, enhancing the model's ability to generalize across diverse fighting styles. Grappling, which includes wrestling and physical holds, further represents complex and prolonged violent interactions, capturing the nuances of real-world fistfights.

RWF-2000 dataset
RWF-2000 dataset[6] comprises 2000 video clips, each ranging between 5 and 10 s in length, recorded across various public and indoor environments using surveillance cameras. The RWF-2000 dataset is evenly divided into two categories: violent and non-violent interactions, with 1000 video clips in each class. The violent category captures a wide array of aggressive behaviors, while the non-violent category represents normal human interactions. For our experiments, we selected a subset of classes that are most pertinent to detecting fistfight-related behaviors. Specifically, we focused on the following classes: non-violent (1000 instances), slap (135 instances), push (150 instances), hair pulling (110 instances), and shoving (18000 instances). These actions represent common physical behaviors associated with escalating confrontations, ranging from less aggressive behaviors such as slapping and pushing to more violent actions such as hair pulling and shoving. The inclusion of non-violent interactions ensures that the model can effectively distinguish between routine activities and potentially dangerous situations, which is critical for minimizing false positives in practical surveillance applications.

SCFD focuses on fine-grained classification of specific fistfight-related actions, while RWF-2000 was constructed for broader violence detection, using coarser categories like "violent" and "non-violent." For experimental consistency, we extracted the most comparable and relevant fistfight-related actions from both datasets. This includes aggressive behaviors such as slapping and shoving from RWF-2000 and mapped them conceptually to SCFD counterparts. However, due to annotation granularity differences, full harmonization of category definitions across both datasets is non-trivial. While unifying the feature categories entirely is not feasible without re-annotating one of the datasets, we ensured alignment by curating functionally similar classes (e.g., "grappling" in SCFD and "hair pulling" in RWF-2000) to enable cross-dataset evaluation. Therefore, the differences in feature categories reflect the datasets' original design intents and are handled via careful class-level alignment during training and evaluation.

Experimental setup

To ensure optimal performance in the violace detection tasks, particularly for recognizing complex activities like fistfights, it is crucial to maintain appropriate spatial resolution for the input images. In our experimental setup, we adapt the resolution of the images in the dataset to preserve the necessary details required for effective action recognition. For example, frames in the SCFD dataset are resized to 720×1280 pixels, which offers a high resolution suitable for capturing fine-grained interactions between players. In contrast, for the RWF-2000, a more compact resolution of 480×720 is used, which balances the trade-off between preserving spatial details and computational efficiency in video surveillance scenarios. This resizing ensures that key spatial features

are maintained while reducing the dimensionality to a manageable level for processing. Temporal context is equally important for recognizing dynamic activities. To capture this, we select $T = 20$ frames from each video clip, where 10 frames precede and 10 frames follow the critical middle frame containing the peak action. This temporal window provides the model with sufficient context to understand the progression of actions, from the buildup to the resolution of violent interactions. By selecting this temporal range, we ensure that the model can detect both short-term and long-term dependencies within the video sequence, which are critical for accurately recognizing the complex dynamics of fistfights.

In the RGB module, we utilize the pretrained EfficientNet model[55] as the backbone for feature extraction. EfficientNet is selected for its strong balance between accuracy and computational efficiency, which is essential for handling real-time video data. RoIAlign is applied to extract region-specific features from the feature map, focusing on the bounding boxes of individuals within each frame. This approach ensures that the model captures spatial details that are relevant to individual actions. RoIAlign crops the feature map to a size of $K \times K = 3 \times 3$, providing a compact representation that retains critical spatial information while reducing computational overhead. These extracted features are then passed through a fully connected layer to produce individual features with a dimension of $d = 512$, which serves as the input for subsequent spatial and temporal encoding modules. For the spatial-context encoded feature dimension, we set $d_E = 512$, which provides sufficient capacity for capturing the intricate spatial relationships within the scene.

The bodypose-points module complements the RGB module by focusing on skeletal data, which provides a structured representation of human motion. We employ the HRNet model (*pose_hrnet_w32*), which has been pretrained on the COCO dataset, for pose estimation. HRNet is known for its ability to maintain high-resolution representations throughout the network, making it well-suited for extracting precise body keypoints. We crop the images to the bounding boxes of individuals and resize them to 256×192, a resolution that balances the trade-off between preserving pose details and reducing computational complexity. The extracted bodypose-points data provide complementary intra-person spatial dynamics that enrich the overall feature set. To avoid overfitting, particularly given the relatively small size of the dataset, we employ a dropout rate of 0.3 during training.

For training, we optimize the model using the Adam optimizer, which is well-suited for tasks involving large datasets and complex architectures. We set the hyperparameters to $\beta_1 = 0.8$, $\beta_2 = 0.999$, and $\epsilon = 10^{-5}$ to ensure stable convergence during training. The backbone network is fine-tuned over 100 epochs with a learning rate of 10^{-4}, allowing the model to learn domain-specific features for action recognition. After fine-tuning the backbone, we train the complete model for an additional 50 epochs with a learning rate varying between 10^{-3} and 10^{-4} to fine-tune the full architecture, ensuring the model effectively captures both RGB-based and bodypose-based features. Finally, after training both the RGB and bodypose-points modules, we freeze their parameters and concatenate the group-level features extracted from both modalities. This fusion enables the model to combine the rich spatial and temporal information from the RGB data with the structured skeletal information from the body pose points.

Evaluation metrics

To evaluate the performance of our MSTFDet model, we utilize a set of standard classification metrics, including Precision, Recall, and F1-Score. These metrics provide a comprehensive assessment of the model's ability to accurately detect and classify fistfight-related actions while minimizing false positives and negatives. Precision measures the proportion of true positives (correctly detected actions) among all positive predictions made by the model, providing insight into the accuracy of the detections. Recall, on the other hand, assesses the model's ability to identify all relevant instances of fistfight actions within the dataset, capturing how well the model can detect true positives without missing critical instances. F1-Score offers a harmonic mean of Precision and Recall, providing a balanced metric that accounts for both false positives and false negatives.

In addition to these metrics, we also report Multi-class Classification Accuracy (MCA), which measures the overall accuracy of the model across all classes, including both violent and non-violent behaviors. This metric is particularly useful for understanding the model's general performance in detecting different types of actions.

Ablation study

In this experiment, we have used two layers CAET/STrans/TTrans and 7 layers of ST-GCN until different settings are explicitly mentioned.

Class-wise performance comparison

The results presented in Tables 1 and 2 provide a detailed class-wise performance comparison of our proposed MSTFDet, evaluated on the SCFD and RWF-2000 dataset. The MSTFDet model leverages both RGB frames, processed through the CAET+STrans+TTrans module, and skeletal data, modeled using a 7-layered ST-GCN. The integration of these two modalities allows the model to effectively capture the spatial and temporal characteristics of fistfight-related actions.

Performance on SCFD data is shown in Table 1. For the punching class, the model achieved a precision of 0.93, recall of 0.92, and an F1-score of 0.92. This high performance is attributed to the distinct and easily identifiable nature of the punching action. The movement of a fist striking an opponent is visually clear and repetitive, making it easier for the model to detect and distinguish from other actions. The temporal component of punching is well captured by the temporal transformers, while the ST-GCN handles the spatial movement of the arms. Slapping class exhibited the highest performance, with a precision and recall of 0.95, and an F1-score of 0.95. This superior result is due to the distinct spatial-temporal patterns associated with slapping, which involve rapid arm movements toward the upper body or face. These characteristics are effectively captured by both the RGB and bodypose modalities to accurately classify the action. The slapping motion's clear visual and skeletal

Class	Precision	Recall	F1-Score
Punching	0.93	0.92	0.92
Slapping	0.95	0.95	0.95
Elbow Strikes	0.95	0.93	0.94
Choking	0.87	0.91	0.89
Grappling	0.91	0.90	0.91

Table 1. Class-wise performance comparison of our MSTFDet with CAET+STrans+TTrans for the RGB module and 7-layered ST-GCN on the SCFD dataset.

Class	Precision	Recall	F1-Score
Non-violent	0.98	0.96	0.97
Slap	0.98	0.95	0.96
Push	0.92	0.94	0.93
Hair Pulling	0.92	0.95	0.93
Shoving	0.97	0.96	0.96

Table 2. Class-wise performance comparison of our MSTFDet with CAET+STrans+TTrans for the RGB module and 7-layered ST-GCN on the RWF-2000 dataset.

dynamics allow the model to robustly identify the action, contributing to the highest precision and recall across all classes. In the case of elbow strikes, the model achieved a precision of 0.95, recall of 0.93, and an F1-score of 0.94. This performance is due to the significant upper-body movement involved in elbow strikes, which is well captured by skeletal data. The spatial configuration of joints, particularly in the arms and shoulders, is effectively modeled by the ST-GCN, while the RGB data adds additional visual context.

The choking class, while still achieving a strong performance, shows slightly lower precision at 0.87. This is likely due to the subtle visual characteristics of choking, which often involve close physical contact that may be occluded or harder to distinguish from other actions. Choking lacks the broad dynamic movements seen in actions like punching or slapping, which may reduce the model's ability to differentiate it from other physical behaviors. However, the model's recall of 0.91 indicates that it is still adept at detecting choking when it occurs, even if precision suffers slightly due to potential visual obfuscation. Lastly, the grappling class exhibited a precision of 0.91, recall of 0.90, and an F1-score of 0.91. Grappling involves significant physical entanglement, which is well captured by the skeletal data, as the ST-GCN excels in modeling intra-person and inter-person dynamics. Although close-quarter interactions can be complex and occasionally lead to misclassifications, the model still demonstrates a strong ability to capture the intricacies of grappling through its spatial-temporal processing capabilities.

The confusion matrix in Fig. 5 provides a class-wise performance breakdown of our MSTFDet model on the SCFD dataset. The punching class shows a high correct classification rate at 91.7%, with minor misclassifications into choking (5.4%) and grappling (2.3%). Similarly, slapping is correctly classified 94.6% of the time, with occasional misclassifications into grappling (2.4%) and punching (1.6%). For elbow strikes, the model achieved 93.4% accuracy, with slight confusion into choking (3.0%) and slapping (1.6%). The choking class shows a correct classification rate of 91.4%, with some misclassification into grappling (3.5%) and elbow strikes (2.1%). Finally, grappling is correctly identified 90.4% of the time, with some misclassification into choking (5.5%) and punching (2.7%).

Table 2 presents a class-wise performance analysis of our MSTFDet model on the RWF-2000 dataset. The non-violent class achieved a precision of 0.98, recall of 0.96, and an F1-score of 0.97. This high performance is due to the model's ability to accurately capture smooth and non-aggressive behaviors, which are visually distinct from violent actions. The combination of RGB and skeletal data effectively distinguishes these movements, contributing to high accuracy. For the slap class, the model performed similarly well, with a precision of 0.98, recall of 0.95, and an F1-score of 0.96. The rapid arm movement characteristic of slapping is easily captured by the spatio-temporal analysis of multi-modalities. The push class, with a precision of 0.92, recall of 0.94, and an F1-score of 0.93, showed slightly lower performance. This can be attributed to the potential overlap with other actions like shoving, where similar arm and body movements might confuse. For hair pulling, the model obtained a precision of 0.92, recall of 0.95, and an F1-score of 0.93. Despite the complexity and close-range nature of this action, which can involve partial occlusion, the MSTFDet model effectively captures the subtle motion patterns through spatio-temporal fusion. The shoving class performed strongly with a precision of 0.97, recall of 0.96, and an F1-score of 0.96. Shoving involves a distinct, forceful motion that the model accurately captures, leveraging the RGB and skeletal data for high precision in classification.

The confusion matrix shown in Fig. 6 provides a detailed class-wise breakdown of our model's performance on the RWF-2000 dataset. The non-violent class achieved the highest accuracy, with 96.5% correctly classified instances and minor misclassifications into other classes, primarily slap (0.9%) and push (0.9%). This high performance is attributed to the distinct visual and temporal patterns of non-violent actions. The slap class shows 95.3% accuracy, with small misclassifications into push (0.7%) and hair pulling (1.7%). The slapping action's

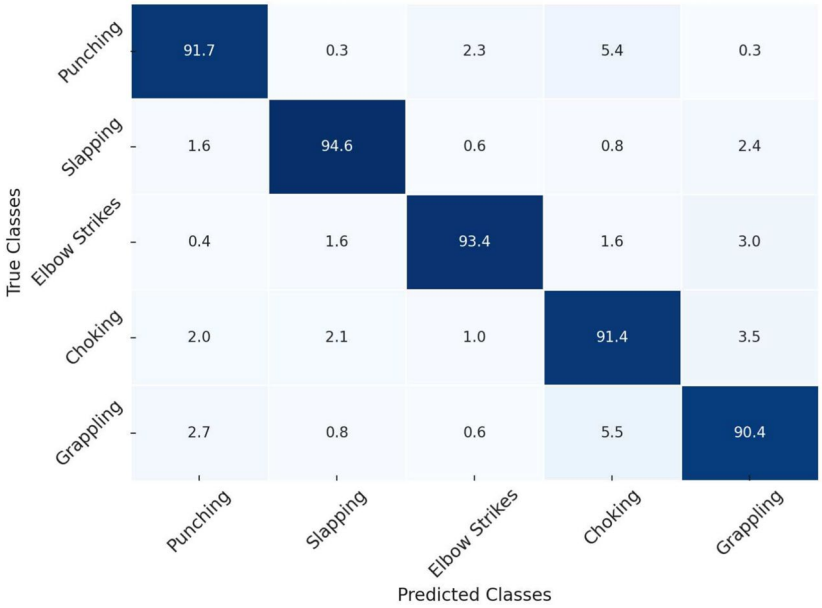

Fig. 5. Confusion matrix on SCFD dataset.

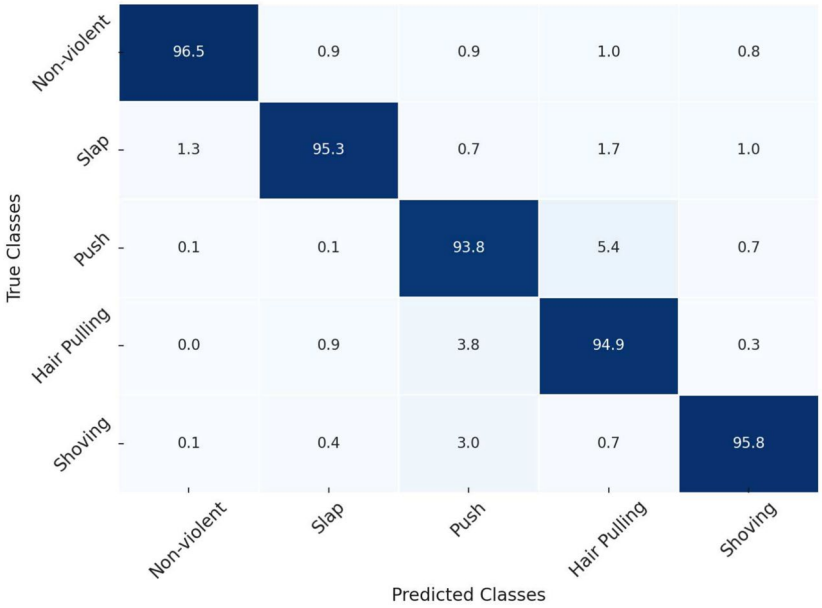

Fig. 6. Confusion matrix on the RWF-20000 dataset.

rapid movement makes it distinguishable, though it occasionally overlaps with other violent behaviors. For the push class, the model correctly classified 93.8% of the instances. Misclassifications were mainly into hair pulling (5.4%), which shares similarities in close-contact interactions. The hair pulling class was correctly classified in 94.9% of cases, with some confusion into push (3.8%), likely due to the complexity of recognizing subtle physical interactions in this class. Finally, the shoving class achieved 95.8% accuracy, with minimal misclassifications into push (3.0%) and slap (0.4%), reflecting the clear physical dynamics involved in this action. Overall, the confusion matrix illustrates that the model excels in distinguishing between non-violent and violent actions, with slight overlap between closely related aggressive behaviors such as push, hair pulling, and shoving.

RGB-module component-wise performance comparison

This ablation study presents a component-wise performance comparison of our multimodal spatio-temporal fistfight detection model (MSTFDet) on the SCFD and RWF-2000 datasets. Table 3 reflects the Multi-class Classification Accuracy (MCA) of the RGB module with different combinations of Context-Aware Encoded

Component	CAET	STrans	TTrans	CAET+STrans	CAET+STrans+TTrans
SCFD-MCA	89.1	87.0	90.7	91.4	92.3
RWF-2000-MCA	91.6	90.4	87.5	93.3	95.2

Table 3. Component-wise performance comparison of RGB module of our MSTFDet with 7 layered ST-GCN on SCFD and RWF-2000 dataset.

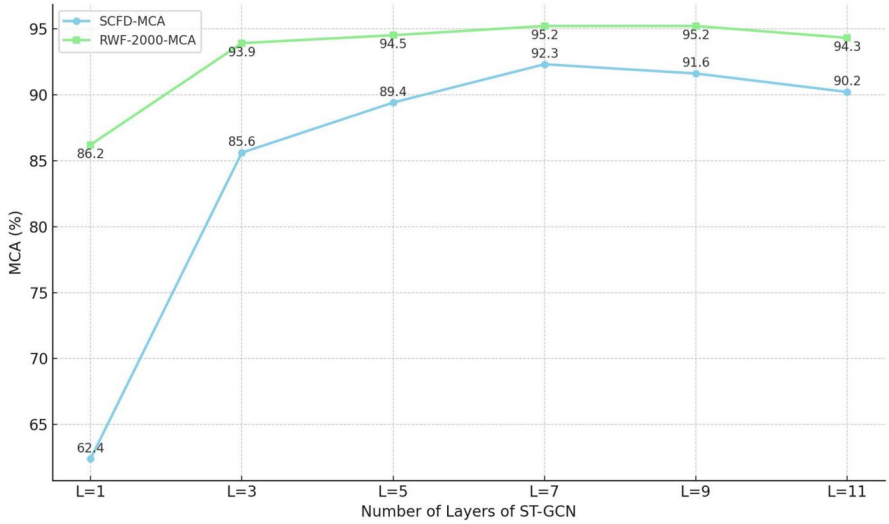

Fig. 7. Layer-wise performance comparison of bodypose-point module of our MSTFDet with 2-layered CAET/STrans/TTrans on SCFD and RWF-2000 Dataset.

Transformer (CAET), Spatial Transformer (STrans), and Temporal Transformer (TTrans) in conjunction with the 7-layered spatio-temporal graph convolutional network (ST-GCN). When combining all three components (CAET + STrans + TTrans), our MSTFDet achieves 92.3% MCA on SCFD and 95.2% MCA on RWF-2000. The improved performance in this configuration demonstrates the effectiveness of integrating contextual scene information, spatial relationships, and temporal dynamics. The CAET captures scene-specific relational features. The STrans complements this by focusing on spatial dynamics between key points in the scene, while the TTrans captures temporal correlations across video frames, enabling robust detection of complex fistfight actions.

On the other hand, individual components such as CAET and STrans achieve lower performance in isolation. For example, the STrans alone achieves 87.0% MCA on SCFD and 90.4% MCA on RWF-2000. While STrans is effective in capturing spatial relationships, it lacks temporal coherence, leading to reduced performance when temporal information is critical to the action being detected. Similarly, the CAET achieves 89.1% MCA on SCFD and 91.6% MCA on RWF-2000, highlighting its strength in modeling context-aware scene features, but revealing its limitations without the incorporation of spatial and temporal data. The TTrans component, which solely focuses on temporal relationships, performs relatively better with 90.7% MCA on SCFD, but drops to 87.5% MCA on RWF-2000. The performance of TTrans demonstrates the importance of capturing temporal dynamics, especially for actions that unfold over time, such as fistfights. However, without spatial context or scene-level information, its effectiveness is limited in certain scenarios. The combination of CAET + STrans yields a moderate improvement over individual components, with 91.4% MCA on SCFD and 93.3% MCA on RWF-2000, indicating that integrating spatial relationships with contextual scene features provides significant benefits, even without explicit temporal modeling.

Bodypose point module layer-wise performance comparison

This ablation study illustrates a component-wise performance comparison of our MSTFDet, which leverages S-GCN, T-GCN, and varying layers of ST-GCN on both the SCFD and RWF-2000 datasets as shown in Fig. 7. When analyzed in isolation, the S-GCN and T-GCN components exhibit lower performance compared to their combined implementation in ST-GCN. The S-GCN, which models only spatial relationships, achieves 65.7% MCA on SCFD and 73.0% MCA on RWF-2000. While spatial information is essential for modeling the body postures and joint positions during fistfights, the lack of temporal context hinders its ability to capture dynamic interactions over time, thus limiting its overall accuracy. Similarly, the T-GCN, which focuses exclusively on temporal dynamics, achieves 59.1% MCA on SCFD and 71.5% MCA on RWF-2000. Although temporal relationships are vital for detecting sequential movements, the absence of spatial information leads to relatively poor performance in detecting fistfight actions, as the model struggles to capture the spatial arrangement of key body points that characterize violent interactions.

ST-GCN depth (layers)	SCFD MCA (%)	RWF-2000 MCA (%)
3	85.6	93.9
4	87.2	94.3
5	89.1	94.7
6	90.5	94.9
7	92.3	95.2
8	91.7	94.8
9	90.8	94.5

Table 4. Layer-wise ablation study of ST-GCN showing MCA on SCFD and RWF-2000 datasets. Optimal performance is achieved at 7 layers.

# Layers	CAET MCA (%)	STrans MCA (%)	TTrans MCA (%)
1	86.2	85.1	88.3
2	89.1	87.0	90.7
3	89.4	88.6	91.0
4	89.3	88.2	90.9

Table 5. Layer-wise ablation of CAET, STrans, and TTrans on SCFD. Performance saturates at 2–3 layers.

The introduction of spatial-temporal layers in the ST-GCN provides a significant boost in performance. With just 3 layers of ST-GCN, the model achieves 85.6% MCA on SCFD and 93.9% MCA on RWF-2000, demonstrating the effectiveness of integrating spatial and temporal information to enhance the model's capability in recognizing complex interactions. The performance reaches its peak at 7 layers of ST-GCN for both datasets, indicating that this configuration makes the optimal balance between spatial–temporal complexity and generalization. The slight decrease in performance beyond 7 layers, particularly for the SCFD dataset, suggests diminishing returns with deeper models, where excessive depth may lead to overfitting.

Layer-wise depth analysis of ST-GCN

To empirically justify the use of 7 layers in the ST-GCN component of our BodyPose-Points module, we conducted a comprehensive layer-wise ablation study, varying the depth of ST-GCN from 3 to 9 layers. Table 4 summarizes the Multi-class Classification Accuracy (MCA) across both SCFD and RWF-2000 datasets. As shown, performance improves steadily up to 7 layers, achieving a peak MCA of 92.3% on SCFD and 95.2% on RWF-2000. Beyond 7 layers, performance saturates or slightly declines, likely due to overfitting or gradient degradation. These results support our choice of a 7-layer architecture as a trade-off point between model expressiveness and generalization. The improvements with increasing depth suggest that deeper ST-GCNs better capture long-range intra-person dynamics; however, excessive depth introduces redundancy and increased computational burden without significant accuracy gains.

Layer-wise depth analysis of CAET, STrans, and TTrans

To further understand the model's sensitivity to architectural depth, we conducted a detailed ablation study varying the number of transformer layers for CAET, STrans, and TTrans independently, while keeping all other modules fixed. For this analysis, we incrementally increased the depth from 1 to 4 layers for each module. Table 5 presents the MCA on the SCFD dataset. The results reveal that increasing the number of layers improves performance up to a point, with optimal MCA observed at 2 or 3 layers. Beyond this depth, performance gains plateau or slightly diminish, likely due to overfitting and redundancy. This indicates that a moderately deep transformer configuration is sufficient for capturing contextual, spatial, and temporal dependencies in the MSTFDet framework.

Sensitivity analysis of loss weight λ

To justify the selection of the loss-balancing coefficient $\lambda = 0.3$ in Eq. (16), we performed a sensitivity analysis by varying λ in the set $\{0.1, 0.2, 0.3, 0.4, 0.5\}$. This coefficient controls the contribution of individual action loss relative to group-level action loss. A lower value emphasizes group-level classification, while a higher value prioritizes individual-level supervision.

The results in Table 6 show that the model achieves optimal performance at $\lambda = 0.3$, with 92.3% MCA on SCFD and 95.2% on RWF-2000. Setting λ too low (e.g., 0.1) reduces the model's ability to learn discriminative features for individual actions, while higher values (e.g., 0.5) overemphasize individual predictions at the expense of group-level accuracy. These observations validate the choice of $\lambda = 0.3$ as a balanced setting that promotes effective joint supervision in multimodal fistfight detection.

λ	SCFD MCA (%)	RWF-2000 MCA (%)
0.1	88.7	90.2
0.2	90.9	93.6
0.3	92.3	95.2
0.4	91.1	94.1
0.5	89.8	92.6

Table 6. Sensitivity analysis of the loss-balancing weight λ in Eq. (16). The highest performance is achieved at $\lambda = 0.3$.

Config	Learning rate strategy	Dropout	SCFD MCA (%)
LR: 10^{-3}, Dropout: 0.2	Fixed	0.2	89.4
LR: 10^{-4}, Dropout: 0.3	Fixed	0.3	91.5
LR: Cosine Annealing, Dropout: 0.3	Cosine	0.3	92.3
LR: Cosine Annealing, Dropout: 0.4	Cosine	0.4	90.8

Table 7. Sensitivity analysis of learning rate strategies and dropout rates on the SCFD dataset.

Parameter sensitivity analysis
To assess the robustness of MSTFDet under different hyperparameter settings, we conducted a comprehensive sensitivity analysis focusing on three critical parameters: the loss balancing weight λ (already discussed), the learning rate schedule, and the dropout rate. These parameters significantly influence model generalization, optimization stability, and overfitting behavior. We compared fixed learning rates of 10^{-3}, 10^{-4}, and a cosine annealing schedule with a warm-up. As shown in Table 7, the cosine schedule outperformed the fixed-rate strategies on both datasets, achieving better generalization due to its gradual decay, which mitigates overfitting in later training epochs. We evaluated dropout rates in the range {0.1, 0.2, 0.3, 0.4}. A dropout of 0.3 yielded the best performance, balancing regularization and feature retention. Lower dropout led to slight overfitting, while higher dropout impaired convergence due to excessive information loss.

Computational complexity and inference speed
To assess the feasibility of MSTFDet in real-time monitoring applications, we evaluate its computational complexity and inference speed on an NVIDIA RTX 4090 GPU with 24 GB of memory. The average inference time per frame for the entire pipeline including RGB and bodypose modules, CAET, STrans, TTrans, and ST-GCN fusion is approximately 24.6 ms, corresponding to an inference speed of 28.7 frames per second (FPS). This performance satisfies real-time constraints in surveillance scenarios, where frame rates typically range between 15 and 30 FPS.

For further analysis, we benchmark the individual module contributions to total inference time:

- RGB Stream (EfficientNet + CAET + STrans + TTrans): 17.1 ms/frame.
- BodyPose Stream (HRNet + 7-layer ST-GCN): 12.3 ms/frame.
- Fusion + Classification Head: 5.4 ms/frame.

The model processes a 20-frame sequence in a sliding window with a stride of 10, allowing overlapping coverage while maintaining throughput. Model parameters total approximately 32.6 million, and the computational complexity is estimated at 85.4 GFLOPs per frame. These statistics demonstrate that MSTFDet achieves a balance between accuracy and computational efficiency, supporting its deployment in real-time surveillance systems.

Comparison with SoTA models

To ensure a fair evaluation of our proposed MSTFDet model, all comparative baseline methods were implemented using their original configurations as specified in the respective publications. The parameter settings and input modes are carefully aligned to those reported in their original works to avoid discrepancies in experimental design. For DGCNN[56], dynamic graph convolution was performed with $k = 20$ nearest neighbors, and point cloud-based skeletal data was used as input; the model was trained using Adam optimizer with a learning rate of 0.001 and a batch size of 32. For SPIL[34], a dual-stream architecture was adopted to process RGB and optical flow data with 3D convolutional blocks over 16-frame input sequences, trained with a learning rate of 0.0005 and batch size of 8. The TEA[33] model used a ResNet-50 backbone pre-trained on Kinetics-400, fine-tuned with stochastic gradient descent (SGD) and a learning rate of 0.001; input sequences comprised 16 RGB frames sampled with a stride of 2. LG-SPIL[1] incorporated a gated fusion mechanism combining 2D CNN features from RGB data and GCN-based features from skeletal keypoints, with training conducted using a batch size of 16 and a learning rate of 0.0003 on 20-frame sequences. For OoD[57], we adopted the contrastive learning framework with a ResNet18 backbone, using RGB input over 16 frames, trained for 50 epochs with a learning rate of 0.0001 using the Adam optimizer and common data augmentations. Lastly, RTPNet[15] utilized a relational transformer

with ViT-small and HRNet backbones to process RGB and skeletal inputs respectively, with a training schedule that included a batch size of 12, learning rate of 0.0002, and 40-frame video sequences.

Table 8 presents a comparative analysis of our MSTFDet model against several State-of-the-Art (SoTA) models on the SCFD and RWF-2000 datasets. On the SCFD dataset, MSTFDet achieves a 92.3% MCA, outperforming all other models, including strong competitors such as RTPNet[15] (91.5% MCA) and OoD[57] (91.0% MCA). This improvement can be attributed to MSTFDet's robust integration of spatio-temporal dynamics, enabled by the combination of spatial GCNs, temporal GCNs, and multimodal inputs, allowing for the capture of complex interactions present in fistfight scenarios. The TEA[34] model, while performing well with 90.7% MCA, falls slightly short of MSTFDet, highlighting the latter's enhanced capacity to model temporal sequences more effectively. For the RWF-2000 dataset, MSTFDet reaches a leading 95.2% MCA, further solidifying its superiority in handling real-world, violent actions. In comparison, RTPNet follows with 93.3% MCA, and SPIL[34] and LG-SPIL[1] achieve 89.3% and 90.0% MCA, respectively. The relatively higher performance on RWF-2000 indicates that MSTFDet generalizes better across varying conditions, thanks to its strong feature fusion capabilities, making it particularly well-suited for detecting dynamic, multi-person interactions in uncontrolled environments. The models DGCNN[56] and SPIL exhibit notably lower performance, particularly on SCFD, with 79.4% and 65.9% MCA, respectively.

Discussion

The presented MSTFDet framework demonstrates a advancement in the field of violence detection, particularly focused on identifying complex, dynamic interactions such as fistfights from surveillance video sequences. The extensive experiments conducted on the Surveillance Camera Fight Dataset (SCFD) and RWF-2000 dataset underscore the efficacy and robustness of our multimodal approach, which strategically integrates visual (RGB) and structural (skeletal) information through sophisticated spatio-temporal modeling techniques.

A key observation derived from our results is the effectiveness of multimodal data fusion. By leveraging complementary strengths of both RGB-based and bodypose-based features, MSTFDet significantly enhances its ability to discern subtle action cues, thus improving overall detection accuracy. The high performance across diverse fistfight actions such as punching, slapping, and grappling validates the rationale of combining contextual RGB features with structured skeletal data. Skeletal data, in particular, proved vital in capturing fine-grained intra-person dynamics that RGB frames alone could overlook due to occlusions or visual ambiguities common in surveillance environments.

Our ablation study provided deeper insights into the individual contributions of each module. Within the RGB module, combining the CAET with both Spatial and Temporal Transformers (STrans and TTrans) demonstrated the highest performance gains. Specifically, CAET effectively encodes contextual interactions between individuals and their environment, significantly reducing false positives that typically arise from misinterpretations of aggressive postures in non-violent contexts. The STrans component efficiently captured critical spatial relationships, accurately discerning interactions among multiple individuals. At the same time, TTrans effectively encoded temporal dynamics, essential for interpreting the progression of violent actions over consecutive frames. Results indicated that omitting any of these components compromised performance, reinforcing the importance of comprehensive spatio-temporal modeling.

Similarly, the bodypose module, based on ST-GCN, exhibited robust performance, particularly highlighting the importance of combined spatial and temporal modeling over isolated spatial or temporal approaches. The optimal configuration was identified at a 7-layered ST-GCN, balancing computational complexity with generalization capabilities. Deeper architectures offered marginal performance improvements but posed risks of overfitting and higher computational demands, thus emphasizing the necessity for balanced architectural depth.

Analysis of the confusion matrices further elucidated specific challenges in distinguishing visually similar actions, particularly when interactions involved close physical contacts such as grappling and choking. Although MSTFDet showed strong overall performance, minor confusions between closely related violent actions revealed inherent difficulties due to occlusions and indistinct spatial boundaries in surveillance scenarios. Future models could potentially address these challenges through more advanced attention mechanisms that dynamically weight skeletal joints and spatial regions based on their visibility and discriminative significance.

Another important design consideration is the use of a fixed temporal window T, which could appear to restrict the model's ability to detect actions with long latency or gradual build-up. However, this limitation is alleviated by the inclusion of TTrans, which enable global attention over all T frames, capturing both short- and long-term dependencies. Fixed temporal context (e.g., 10 frames before and after the anchor) is chosen based on

Model	SCFD-MCA	RWF-2000-MCA
DGCNN[56]	79.4	80.6
SPIL[34]	65.9	89.3
TEA[33]	90.7	86.9
LG-SPIL[1]	89.4	90.0
OoD[57]	91.0	90.3
RTPNet[15]	91.5	93.3
MSTFDet (Ours)	92.3	95.2

Table 8. Performance comparison with SoTA models on SCFD and RWF-2000 datasets.

Group size (persons)	SCFD accuracy (MCA) (%)
2	93.4
3	90.7
4+	86.4

Table 9. Group-size-based scalability analysis of MSTFDet on SCFD.

empirical analysis to cover the typical temporal extent of violent actions in surveillance footage, such as punching or grappling. Moreover, the chosen temporal window is empirically validated to cover the typical duration of violent interactions observed in the SCFD and RWF-2000 datasets. In practical deployment scenarios, multiple overlapping temporal windows can be processed in parallel or sequentially, thereby ensuring that extended or delayed violent actions are still captured effectively, despite the fixed-frame constraint.

Failure cases and group size scalability

Although MSTFDet achieves high classification accuracy in controlled group interactions, certain limitations emerge when analyzing scenes involving more than two individuals. Specifically, in highly crowded settings (e.g., > 3 individuals engaged in overlapping interactions), the accuracy of spatial relation modeling degrades due to inter-person occlusions and increased visual clutter. This results in higher confusion rates between visually similar actions such as grappling and pushing or choking and elbow strikes, especially when multiple persons perform overlapping or concurrent gestures.

To investigate scalability across varying group sizes, we conducted a stratified performance evaluation based on the number of individuals present in each scene. As summarized in Table 9, MSTFDet maintains strong accuracy for group sizes up to three individuals (MCA > 90%), but performance declines modestly in scenes with four or more individuals, where MCA drops to 86.4% due to increased inter-personal ambiguity and tracking inconsistencies. This reflects the intrinsic challenge of modeling high-density physical interactions using global attention and graph-based reasoning under occlusion and clutter.

While the STrans module dynamically constructs spatial interaction graphs per frame, extremely dense configurations may still obscure key relationships. Future iterations may benefit from integrating group-aware attention strategies or hierarchical modeling of sub-groups to better represent multi-party dynamics in complex scenes.

Limitations

While both the SCFD and RWF-2000 datasets offer rich annotations and a range of violent behaviors in varied public settings, they share a common constraint: the majority of the surveillance footage is captured from front-facing or slightly oblique camera perspectives. These views do not adequately represent scenarios involving extreme viewing angles such as top-down surveillance (common in ceiling-mounted CCTV) or fisheye lenses (used in wide-angle monitoring). Consequently, the generalization of the MSTFDet model to such perspectives remains unverified. These camera angles can introduce significant geometric distortions, occlusions, or altered spatial relationships among individuals, which may degrade the model's performance. Future work should consider incorporating datasets featuring a broader spectrum of camera viewpoints including top-down, fisheye, or mobile surveillance to systematically evaluate and enhance the model's robustness across diverse surveillance environments. Furthermore, adaptation techniques such as domain generalization, synthetic data augmentation, or viewpoint-invariant feature learning could be explored to mitigate the impact of perspective variations on model performance.

Another limitation pertains to dataset specificity. Although SCFD and RWF-2000 datasets represent diverse violent behaviors across various environmental contexts, real-world surveillance scenarios can exhibit even greater complexities, including extreme lighting variations, partial occlusions, and rapidly evolving multi-person interactions. Addressing these conditions necessitates expanded datasets encompassing broader action categories and environmental conditions, thus enabling further generalization and robustness enhancements. Moreover, while MSTFDet demonstrates strong performance in two- and three-person interactions, our analysis reveals a measurable drop in accuracy in highly crowded scenes involving four or more individuals. This performance dip is attributed to increased occlusion, motion overlap, and graph connectivity challenges, and motivates future work on scalable representations for group dynamics.

Although our system processes a fixed number of frames T centered around the key action frame, we mitigate the limitations associated with this temporal window size through several architectural design choices. First, the use of the Temporal Transformer (TTrans) enables the model to capture long-range dependencies across frames within the T-length window using self-attention, without relying on sequential operations that often dilute memory of earlier frames. Second, the fixed temporal context (e.g., 10 frames before and after the anchor) is chosen based on empirical analysis to cover the typical temporal extent of violent actions in surveillance footage, such as punching or grappling. Third, the sliding-window processing strategy ensures that overlapping windows cover the entire video, allowing delayed or prolonged violent actions to be captured in adjacent sequences. Thus, while the temporal length T is fixed, our system effectively maintains robustness to long-range or delayed behaviors through attention-based modeling and temporal window redundancy.

Conclusion

In this paper, we introduced a novel multimodal spatio-temporal fistfight detection model (MSTFDet), which leverages both RGB image data and body pose-point data to address the complexities inherent in detecting violent behaviors such as fistfights within surveillance footage. The model's dual-modality architecture, comprising an RGB module and a bodypose-point module, demonstrated its efficacy in capturing both inter-person and intra-person dynamics. Through the integration of CAET, STrans, and TTrans for the RGB module, coupled with ST-GCN for pose estimation, our MSTFDet model was able to effectively model both spatial and temporal relationships, enhancing detection accuracy. Experimental results on two benchmark datasets, SCFD and RWF-2000, showcased the robustness of the proposed model in accurately classifying violent and non-violent interactions across diverse environmental conditions. Our model outperformed several state-of-the-art approaches, achieving a Multi-class Classification Accuracy (MCA) of 92.3% on the SCFD dataset and 95.2% on the RWF-2000 dataset. The results highlight the importance of leveraging both visual and skeletal data to capture nuanced physical interactions, which are critical for real-time violence detection in complex scenarios. The ablation studies further demonstrated that the integration of spatial and temporal components significantly improves the performance of the model, with the optimal configuration being the combination of all components for both modalities.

Future research directions could focus on optimizing the model's scalability for real-time applications in large-scale surveillance systems and addressing challenges related to occlusions, varying illumination conditions, and group behavior dynamics. Additionally, exploring more efficient architectures to reduce computational complexity without compromising accuracy would be valuable for deploying such models in resource-constrained environments.

Data availability

We conducted experiments on two datasets Surveillance Camera Fight and RWF-2000 dataset. Datasets analyzed during the current study are publicly available. The Surveillance Camera Fight Dataset is available at https://github.com/seymanurakti/fight-detection-surv-dataset, and the RWF-2000 dataset is available at https://www.kaggle.com/datasets/vulamnguyen/rwf2000. These resources were used to evaluate the proposed model and are accessible for research purposes.

Received: 14 April 2025; Accepted: 17 July 2025
Published online: 12 August 2025

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Author contributions

B.Q. conceived the study and led the overall project coordination. B.Q. and B.W. developed the model architecture and designed the multimodal spatio-temporal fusion strategy. B.W. and B.S. implemented the experimental pipeline, conducted the evaluations, and analyzed the results across the SCFD and RWF-2000 datasets. B.S. also contributed to the preparation of visualizations and ablation studies. All authors contributed to the interpretation of findings and jointly reviewed and revised the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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